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University Internship Report

DEEP LEARNING FOR PSYCHOGRAPHIC RECONSTRUCTION: ARCHITECTURE BENCHMARKING AND THE CRITICAL ROLE OF ACQUISITION DESIGN

Architecture Experiments, Findings, and
Dataset-Design Recommendations

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September 2026

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Internship period: July – August 2026

Engineering

Abstract

Ptychography is a technique that reconstructs a sample’s amplitude and phase from overlapping diffraction patterns. Classical iterative solvers do this well, but they are too slow for real-time imaging. Deep learning models can fix this by predicting the amplitude and phase directly from the diffraction patterns, speeding up the process by hundreds of times. During my internship, I tested four neural network architectures (PtychoNN, PtychoNN with skip connections, PtychoPINN, and W1-Net) using data from an in-house visible-light optical setup. I tested these on two physical patterns: a Siemens star and an LED array. The results were mixed but highly informative. On the regular, grid-like LED array, the W1-Net model performed exceptionally well, achieving a sharp phase reconstruction ($L_1 = 0.1286$). However, on the complex Siemens star, all models hit a performance wall. I found that this bottleneck was caused by the physical dataset itself, not the models. Specifically, using a large defocused $\sim 300\text{ }\mu\text{m}$ laser probe with a small $60\text{ }\mu\text{m}$ step size created an unintended $\sim 88\%$ scan overlap. This caused a 92% similarity across all training samples, depriving the networks of the feature diversity they need to learn. Finally, I propose specific hardware adjustments and a synthetic pretraining strategy to overcome these data limits in the future.

1 Introduction

Ptychography is a computational imaging technique that creates high-resolution images by processing numerous diffraction patterns, which are recorded as a sample is scanned through a localized illumination beam. This method offers three major advantages: it does not require a physical lens, it can determine both the phase and intensity of the image, and it can self-calibrate to remove experimental errors. While historically considered a form of Coherent Diffractive Imaging (CDI), ptychography is distinct; conventional CDI typically uses a single diffraction pattern to recover an image, whereas ptychography’s core principle is the use of multiple, overlapping diffraction patterns to ensure a highly robust reconstruction.

The fundamental challenge in this technique is known as the "phase problem." The diffraction pattern captured by the camera sensor only records the square root of the amplitude of the sample’s intensity. This represents the modulus of the sample’s transmittance, where 1 equals total transmittance (free space) and 0 equals total absorption. However, when light passes through a sample, the phase of the exit wave is shifted based on the physical depth and structure of the object. This phase information is absolutely crucial for the full reconstruction of the image, but it cannot be physically captured by a camera.

To solve this phase problem, we must use a phase retrieval algorithm. Traditional iterative solvers like ePIE [1] handle this well by mathematically looping back and forth to guess the missing phase, but they require hundreds of computational cycles per image. This makes them far too slow for real-time applications.

To solve this speed issue, deep-learning models were introduced. As illustrated in Figure 1, the convolutional neural network acts as a direct end-to-end mapper: it takes a single far-field diffraction pattern as input through a shared encoder, which then branches into two parallel decoder pathways to predict the sample’s real-space amplitude and phase simultaneously, bypassing the slow iterative loops entirely.

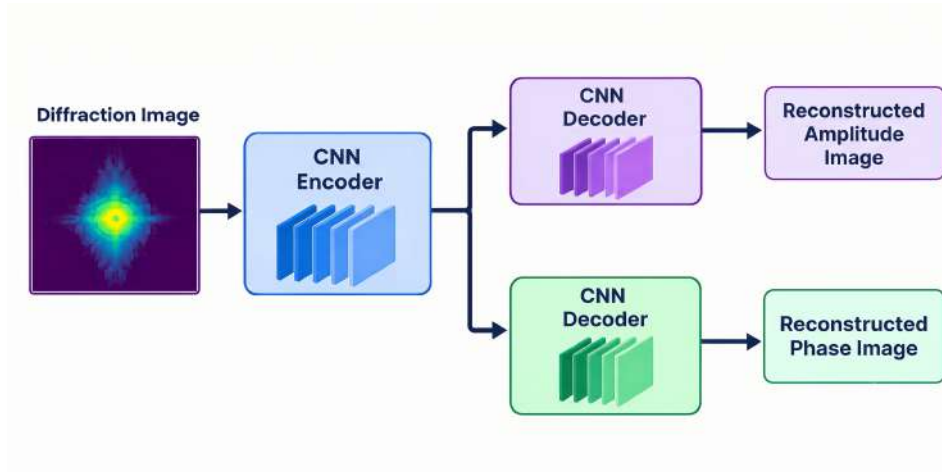


Figure 1: Conceptual overview of deep learning-based model

1.1 Goal of This Internship

The goal of this internship was to build upon the deep-learning ptychography approach using an in-house visible-light optical setup, rather than a traditional X-ray setup. My specific goals were to:

- build a software pipeline to convert raw camera diffraction images and completed ePIE reconstructions into properly formatted training data for neural networks;
- implement and train several candidate architectures across different physical test patterns;
- evaluate and honestly compare their performance; and
- identify the root causes whenever results fell short of expectations—determining whether the fault was in the AI models or the physical hardware setup—and recommend concrete next steps.

This report documents the dataset, the architectures I tested, the benchmark experiments, and my final findings. I specifically examine why the models succeeded on simple patterns but struggled on complex ones due to physical acquisition limits.

2 Related Work

Four models anchor the work in this report.

2.1 PtychoNN

Cherukara et al. [2] introduced the original PtychoNN architecture. It uses a single shared encoder that splits into two decoders (one predicting amplitude and one predicting phase). It learns by directly comparing its predictions to ground-truth images. I used this as the baseline model for the project.

2.2 PtychoNN with Skip Connections

To improve the baseline PtychoNN, I added skip (residual) connections between the matching encoder and decoder layers. This is a standard deep learning technique that helps pass fine details directly to the output, preventing the reconstructed images from looking blurry.

2.3 PtychoPINN

Hoidn et al. [3] created PtychoPINN, a physics-informed variant. Instead of only relying on pre-calculated ground-truth labels, PtychoPINN adds a physics simulation to the training loop. The network takes its predicted amplitude and phase, multiplies them by the known laser beam shape, applies a Fourier transform to simulate diffraction, and then compares that simulated diffraction to the real camera data. This forces the network to obey optical physics rather than just memorizing images.

2.4 W1-Net

Xing et al. [4] proposed W1-Net, a deeper successor to PtychoNN. It is built using modern "ConvNeXt" blocks and attention mechanisms that help the network focus on important spatial frequencies. It was designed to achieve better performance with less data. I rebuilt W1-Net from the descriptions in the published paper.

3 Dataset

3.1 Physical Samples and Experimental Acquisition

The data for this project was collected using an optical ptychography setup with a 632.8nm Helium-Neon (HeNe) laser. The samples were graphene and PMMA thin films placed on a silicon substrate, featuring two main patterns: a Siemens star and an LED array test chip.

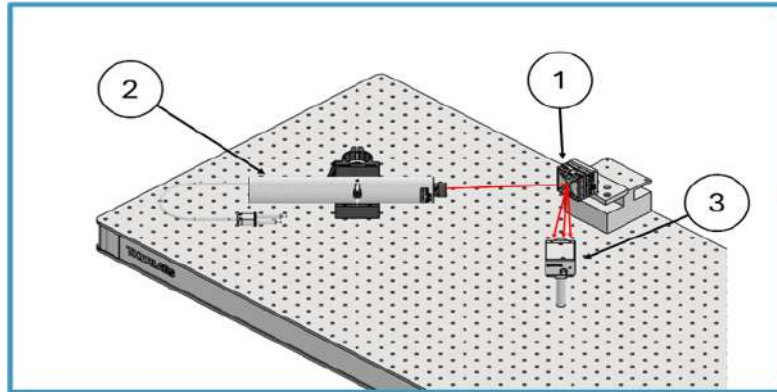


Figure 2: Experimental setup for acquiring diffraction images.

As shown in Figure 2, the hardware included a piezo actuator stage to physically scan the sample (1), the HeNe laser (2), and a 16-bit monochromatic camera sensor to take pictures of the diffraction patterns (3). Data was collected across several different sessions, which introduced variations in laser alignment, lens choices, and camera angles.

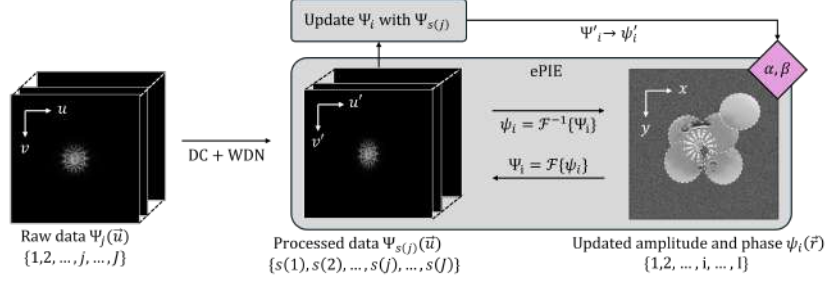


Figure 3: Phase iterative engine (ePIE) workflow used for ground-truth recovery.

Before this internship began, the raw camera images were cleaned to remove background noise and sensor artifacts. To generate the "ground truth" labels that my neural networks would learn from, the data was run through the ePIE algorithm. As shown in Figure 3, ePIE works by looping back and forth between real space and simulated Fourier space, gradually updating its guess until the simulated diffraction matches the camera data.

3.2 What This Internship Did With the Data

My first major task was building a data pipeline to pair the raw diffraction images with their corresponding ePIE ground-truth patches. Because the different experiments were captured with varying camera zooms (different pixel resolutions), I could not just crop a standard 256×256 pixel box. Instead, I scaled the crops based on actual real-world physical sizes, ensuring that one pixel in the neural network always represents the exact same physical distance regardless of which experiment folder it came from.

I also applied a logarithmic transformation ($\log(1+x)$) to the diffraction data to compress the extreme brightness of the laser, mapping the values neatly between 0 and 1 using the camera's 16-bit limit. Figures 4 and 5 show examples of these extracted data triplets (the diffraction input, and the target amplitude and phase). Finally, I shuffled these patches into unified datasets for the neural networks to train on.

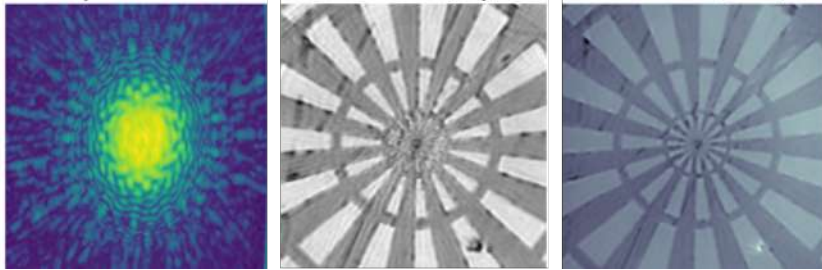


Figure 4: Representative matched patch triplet for the Siemens star pattern.

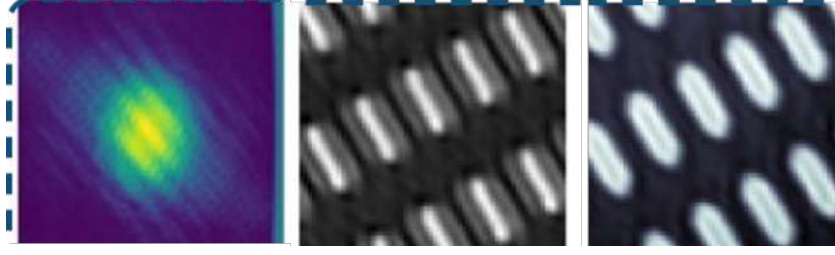


Figure 5: Representative matched patch triplet for the LED array pattern.

4 Methodology

4.1 Architectures

4.1.1 PtychoNN (Baseline)

The baseline PtychoNN model maps diffraction patterns straight to the real-space object properties. As shown in Figure 6, a shared encoder shrinks the image down to extract features. It then splits into two decoders. The amplitude decoder uses a Sigmoid activation to keep pixel values between 0 and 1, while the phase decoder uses a Tanh activation (scaled by π) to keep values between $-\pi$ and π .

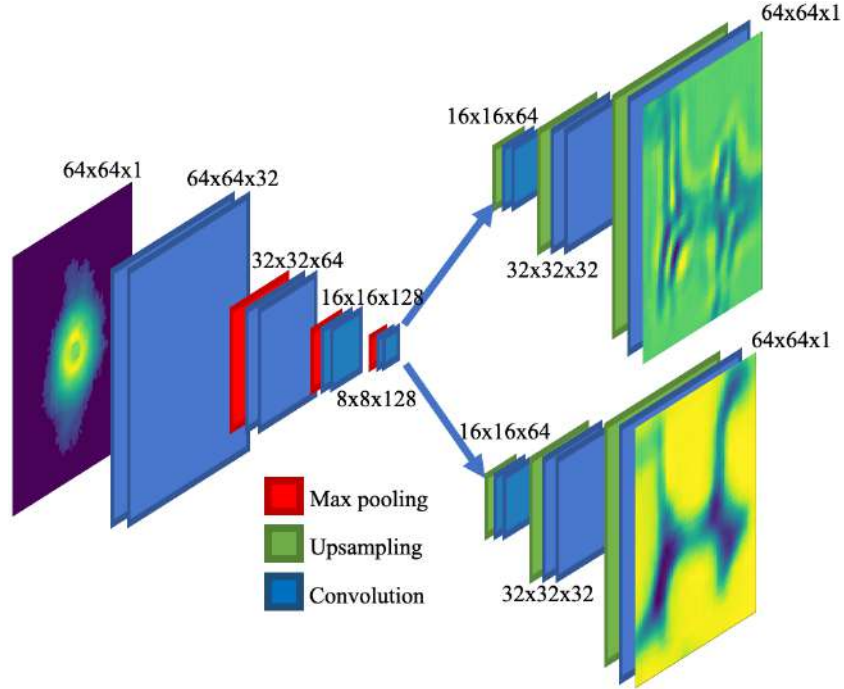


Figure 6: PtychoNN baseline architecture.

4.1.2 PtychoNN with Skip Connections

Every time an encoder shrinks an image, some fine details are lost. To fix this, I added skip connections to the baseline model. These connections take the high-resolution features from the early encoder stages and paste them directly into the decoder stages, helping the network draw sharper boundaries.

4.1.3 PtychoPINN

As shown in Figure 7, PtychoPINN adds a physical forward model to the training process. It takes the network’s predicted amplitude and phase, multiplies it by the laser probe, and uses a Fast Fourier Transform (FFT) to simulate a diffraction image. It calculates its error by comparing this simulation to the real camera data using a Poisson loss function, essentially verifying its own work using optical physics.

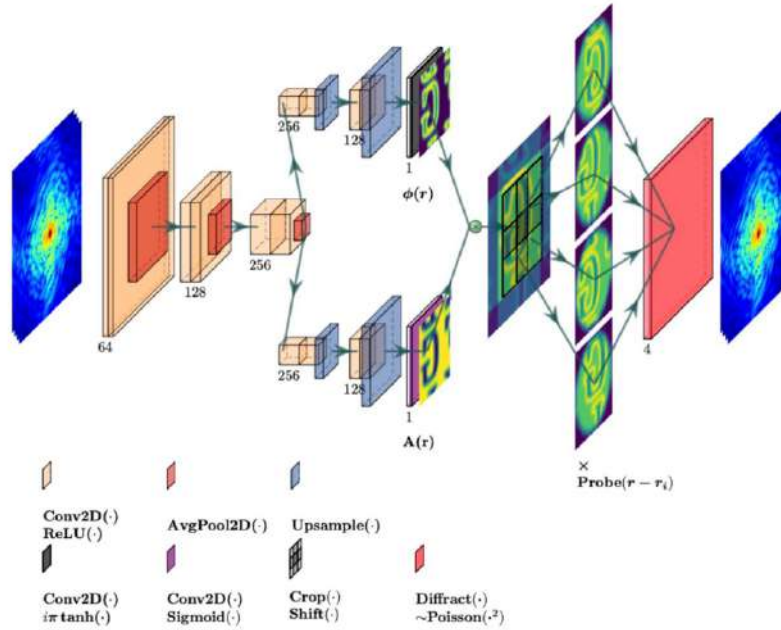


Figure 7: PtychoPINN architecture with forward-model loss.

4.1.4 W1-Net

W1-Net uses advanced "ConvNeXt" blocks instead of standard convolutions, and it includes "Squeeze-and-Excitation" attention modules. This attention mechanism helps the network figure out which spatial frequencies are most important, allowing it to recover sharp images more reliably than the standard PtychoNN.

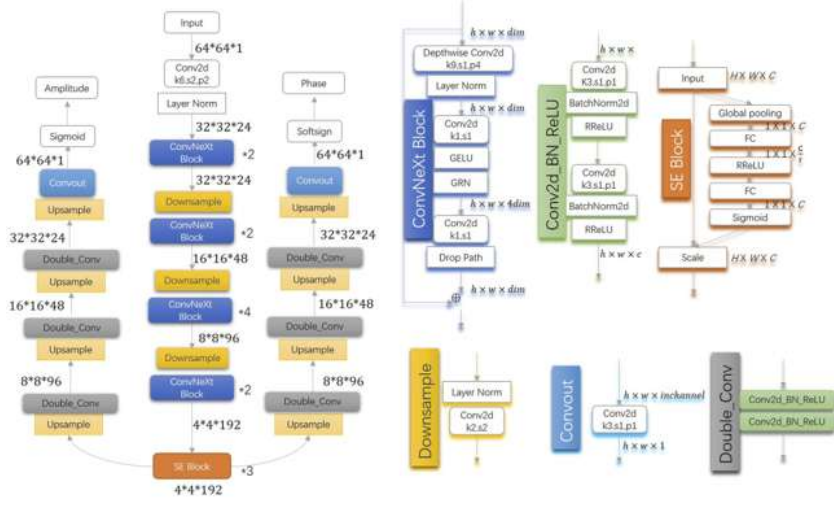


Figure 8: W1-Net architecture with ConvNeXt and SE attention.

4.2 Training Setup

I trained the models using the Adam optimizer and an L1 loss function (Mean Absolute Error) to compare the predicted amplitude and phase against the ground truth. The physics-informed variant also added a Poisson loss term.

4.3 Pipeline Verification on Benchmark Data

Before training the models on our in-house laser data, I needed to make sure my code actually worked. I tested the pipeline using the public X-ray ptychography dataset provided by the original PtychoNN authors (Cherukara et al.). Using only 800 training samples, the network trained in under a minute and successfully reconstructed the sample (Figure 9). itself, not my programming.

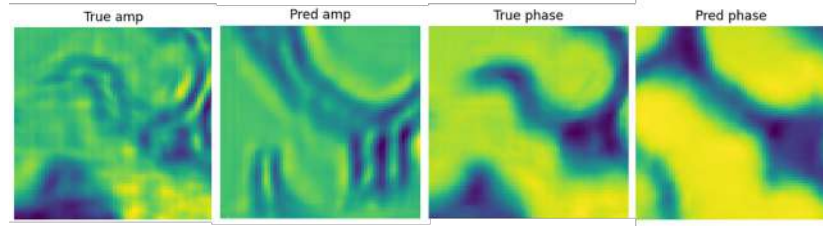


Figure 9: Pipeline verification: PtychoNN reconstruction on 800 samples of public synchrotron X-ray data.

5 Experiments and Results

I ran eight main experiments in total, testing the four architectures on both the Siemens star and the LED array datasets. A notable methodological difference across these runs was the input image resolution. For the first three models (PtychoNN, Skip Connection, and PtychoPINN),

the input diffraction and object patches were sized at 224×224 pixels. Because these networks are fully convolutional, they can smoothly handle variable input dimensions, and they performed well at this higher resolution. However, for W1-Net, the input was explicitly set to 64×64 pixels. Empirical testing revealed an architectural sensitivity: while the simpler baseline models thrived at 224 pixels, W1-Net’s performance degraded at the larger size and proved far more effective and stable at the 64×64 resolution.

As summarized in Tables 1 and 2, the overall results show a very clear pattern: the performance depended heavily on which dataset was used, rather than which model was used.

Table 1: Experiment Results: Siemens star Pattern (Train: 763, Val: 84, Test: 123)

Architecture	Final Test Loss	Amp (L1)	Phase (L1)
PtychoNN	0.5525	0.1455	0.4070
PtychoNN with Skip Connection	0.5582	0.1514	0.4068
PtychoPINN	0.5697	0.1469	0.4229
W1-Net	0.5789	0.1804	0.3985

Table 2: Experiment Results: LEDS Pattern (Train: 832, Val: 92, Test: 132)

Architecture	Final Test Loss	Amp (L1)	Phase (L1)
PtychoNN	0.3562	0.0719	0.2843
PtychoNN with Skip Connection	0.3628	0.0718	0.2910
PtychoPINN	0.3666	0.0736	0.2930
W1-Net (200 Epochs)	0.2089	0.0803	0.1286

To understand the underlying training dynamics, I looked closely at the loss curves. For the Siemens star (Figure 10), the models learned quickly for the first 10 epochs and then completely stopped improving. Because the learning clearly flattened out so early, training for these baseline models was stopped at 60 epochs. Extended testing confirmed this decision: running the models to 200 epochs resulted in a validation loss of 0.4769, which is barely better than the 60-epoch loss of 0.4806. This amounts to a negligible 0.8% improvement, proving that further training was a waste of computational resources. Even the more advanced W1-Net hit a wall on the Siemens star around epoch 36, after which the validation error stabilized and started drifting slightly upward, indicating mild over-specialization to the training data. Figure 11 shows the test predictions: the models recovered the basic amplitude shapes, but struggled badly with the phase, which caused the majority of the error.

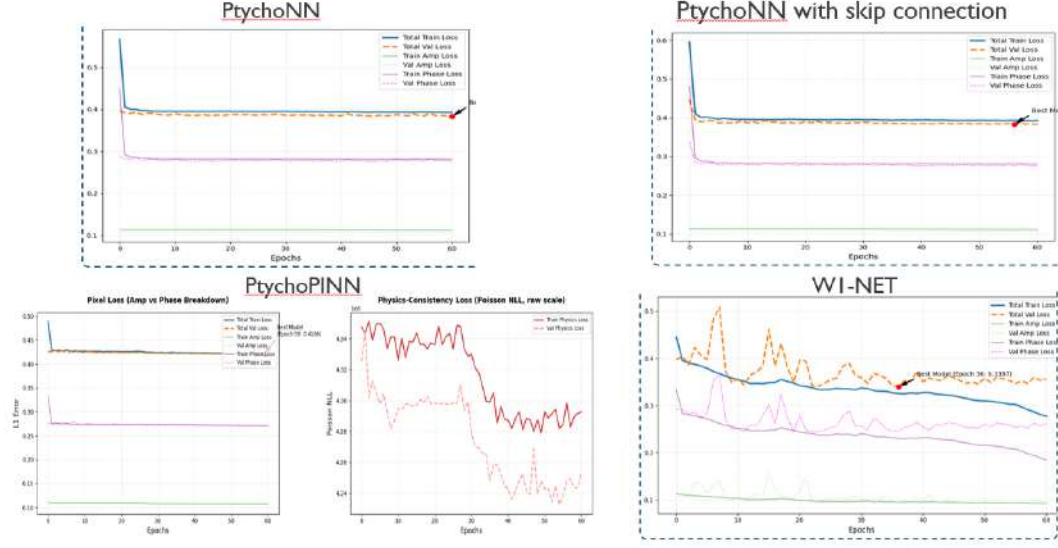


Figure 10: Loss curves for the Siemens star pattern across all four architectures.

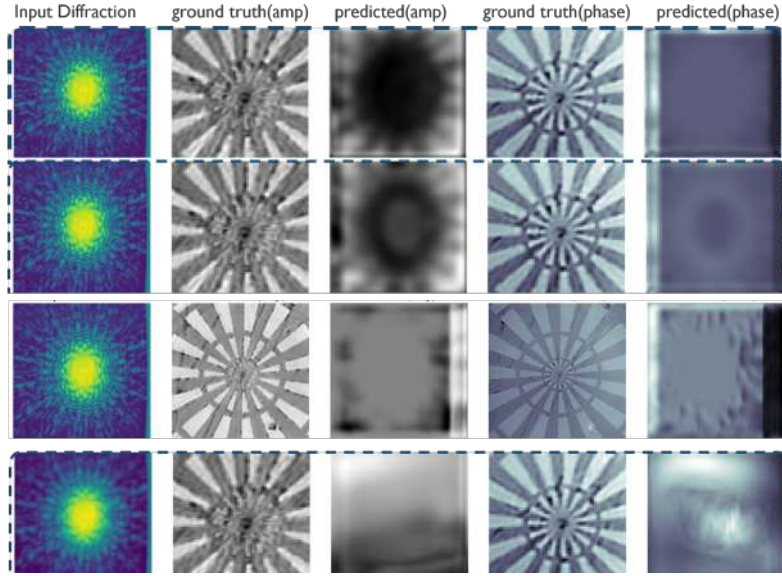


Figure 11: Final test predictions for the Siemens star pattern across the four models.

The LED array dataset yielded completely different results. Unlike the earlier models that stalled, the W1-Net validation curve on the LED data was still sloping downward and actively converging at epoch 60. It showed absolutely zero signs of classic overfitting. Because the model was clearly still learning, I deliberately increased the training duration to 200 epochs. As illustrated in Figure 12, this extended training allowed the loss to keep dropping smoothly and steadily. Because the LED pattern consists of repeating, grid-like structures, the W1-Net was able to learn it perfectly without diverging. As seen in Figure 13, the final predictions on unseen

LED test data were incredibly sharp, heavily reducing the phase error down to 0.1286.

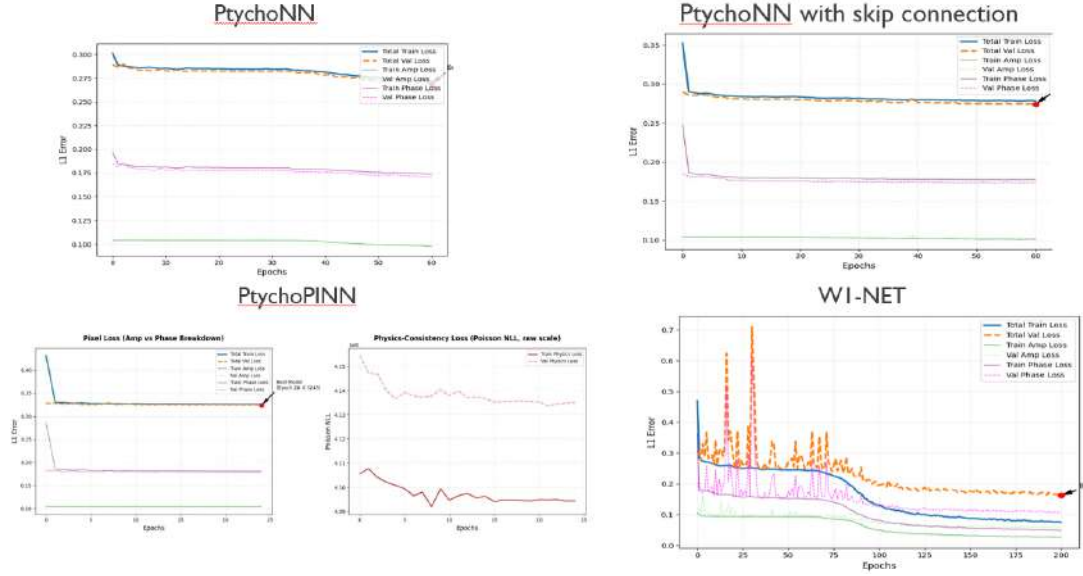


Figure 12: Loss curves for the LED pattern across all four architectures.

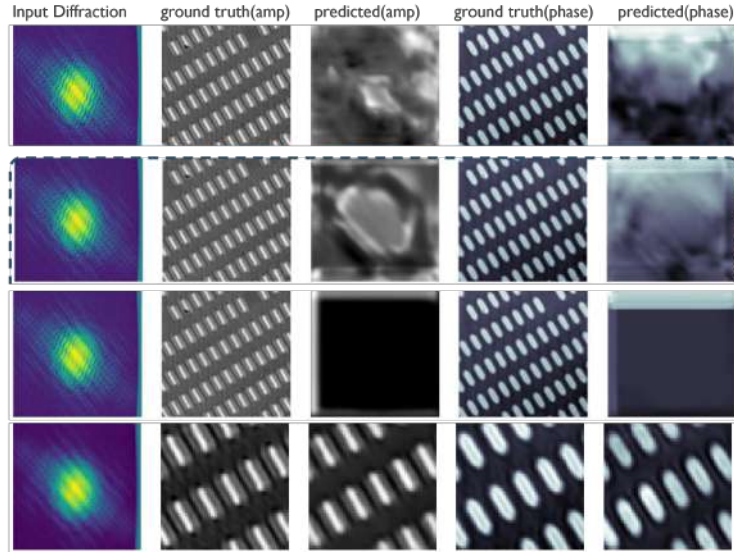


Figure 13: Final test predictions for the LED pattern across the four models. W1-Net (bottom row) achieves highly accurate, sharp phase reconstructions.

Figure 14 shows what these small patch predictions look like when stitched together into a full field of view. The main takeaway from these experiments is that W1-Net performs brilliantly on the LED pattern, breaking through the blurry phase issues that affected the simpler baseline models.

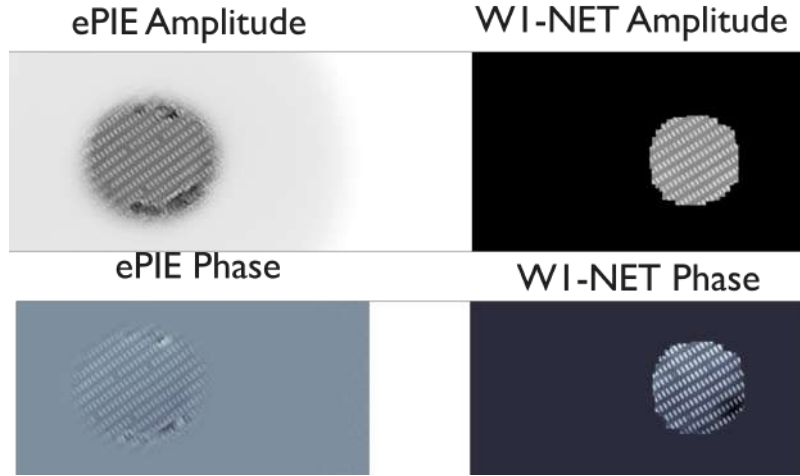


Figure 14: Full aggregate reconstruction of the object (Ground Truth vs. W1-Net prediction).

6 Discussion: Why the Siemens Star Failed

I found three physical causes that explain why all the models struggled with the complex Siemens star pattern.

6.1 Inconsistent Ground Truth Quality

The raw data came from several different experiments with slightly different laser alignments and lenses. Consequently, the ePIE "ground truth" reconstructions varied significantly in quality, as seen in Figure 15. Some folders had sharp images, while others had localized artifacts or blurry phase contrast. Feeding these inconsistent labels into a single training pool confused the neural networks, creating noise that no architecture could learn through.



Figure 15: Ground-truth reconstruction quality across different acquisition folders, showing significant variations in sharpness and artifact levels.

6.2 A 92% Correlation Floor

When I analyzed the training patches, I found that even the most visually different samples were mathematically 92% identical to each other. The data was simply too repetitive. While W1-Net

could handle the repeating lines of the LED array, this lack of diversity made it impossible for the models to learn the complex curves of the Siemens star. The models plateaued early because they weren't seeing enough distinct features to learn anything new.

6.3 The Root Cause: Large Probe, Small Step Size

This lack of diversity was directly caused by the physical hardware scan settings. In the original X-ray paper, the researchers used a tiny 60 nm beam and moved it in 30 nm steps to get a clean 50% overlap. In our visible-light lab, the laser probe was roughly $\sim 300\mu\text{m}$ wide. However, the mechanical stage was only moving 60 μm at a time.

Because the stage step size was so small compared to the massive laser beam, the spatial overlap between shots accidentally ballooned to $\sim 88\%$. Practically, this means the laser barely moved between pictures, so the camera was essentially taking the same picture over and over again. This extreme overlap is what caused the 92% similarity floor and starved the neural networks of diverse training data.

7 Conclusion

This internship successfully established a full deep-learning framework for an in-house visible-light ptychography system, covering everything from raw data extraction to model evaluation.

The results highlight two main takeaways:

- For simple, repeating structures (like the LED array), having a powerful network matters. W1-Net trained for 200 epochs successfully recovered sharp phase contrast ($L_1 = 0.1286$) where older models failed.
- For complex structures (like the Siemens star), the physical hardware setup is the ultimate bottleneck. All models failed to reconstruct the phase properly because the massively defocused $\sim 300\mu\text{m}$ probe caused an $\sim 88\%$ scan overlap. This repetitive data capped the performance across the board.

These findings show that while advanced AI models can do amazing things, they are completely dependent on smart, hardware-aware data acquisition.

8 Future Work

To get better results on complex patterns in the future, we need to fix how the physical data is captured. I recommend the following practical steps:

8.1 Experimental Redesign

Before running new experiments, the scanning parameters must be adjusted:

1. **Measure the probe first:** We need to measure the exact size of the laser spot hitting the sample before planning the scan trajectory.
2. **Adjust the step size for 50% overlap:** The stage step size must be set to approximately half the diameter of the measured laser spot, avoiding the extreme 88% overlap.
3. **Use hardware binning:** Instead of resizing the images in software, we can use the camera's hardware "binning" feature to group pixels together during capture. This naturally shrinks the image and improves the signal-to-noise ratio.

8.2 Synthetic Pretraining

Instead of relying entirely on physical lab data, we can write a Python script to simulate diffraction patterns by passing our measured laser probe over thousands of computer-generated shapes. We can pretrain our models on this massive synthetic dataset, and then fine-tune them on the small amount of real lab data. This would bypass the diversity limits of our physical setup.

References

- [1] A. M. Maiden and J. M. Rodenburg, “An improved ptychographical phase retrieval algorithm for diffractive imaging,” *Ultramicroscopy*, vol. 109, pp. 1256–1262, 2009.
- [2] M. J. Cherukara, et al., “AI-enabled high-resolution scanning coherent diffraction imaging,” *Applied Physics Letters*, vol. 117, 2020.
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- [4] C. Xing, et al., “W1-Net: a highly scalable ptychography convolutional neural network,” *Advanced Optical Technologies*, vol. 13, 2024.